



AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

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4.Create an EC2 Linux Instance in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

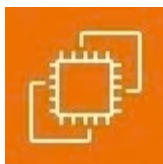
Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

Development and Testing: Create development environments with linux OS to test php, MySQL or other Linux-based applications.

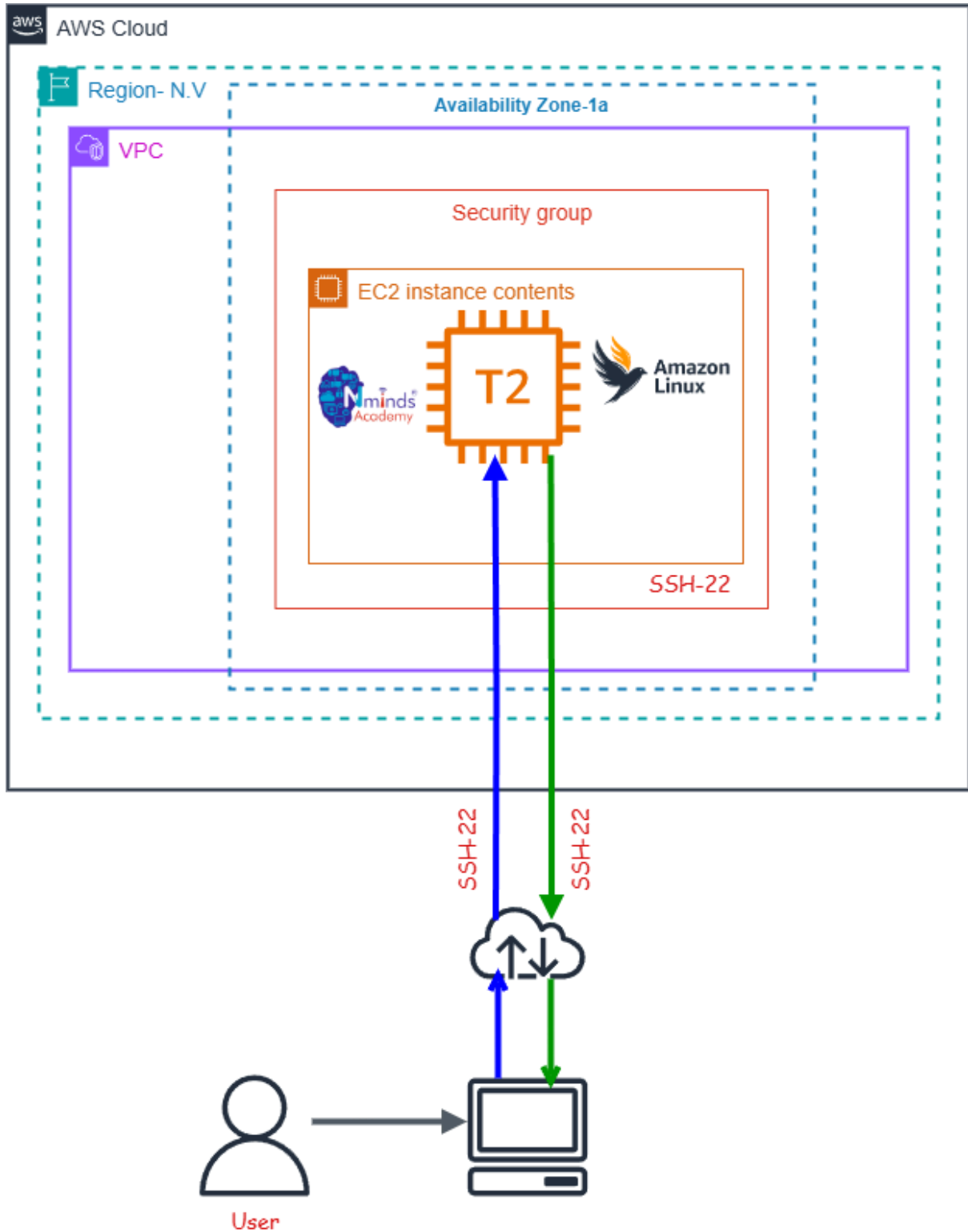
Used Services in AWS



Amazon EC2



Topology



Execution Tasks:

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The top navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The breadcrumb trail is 'EC2 > Instances > Launch an instance'. A search bar at the top of the main content area says 'Search our full catalog including 1000s of application and OS images'. Below this is a 'Quick Start' section with buttons for various operating systems: Amazon Linux, macOS, Ubuntu, Windows (selected), Red Hat, SUSE Linux, and Debian. To the right of these buttons is a 'Browse more AMIs' link. Below the 'Quick Start' section is the 'Amazon Machine Image (AMI)' section, which displays the 'Microsoft Windows Server 2022 Base' AMI (ami-0f7a0c94dce9ab456). It includes details like 'Virtualization: hvm', 'ENA enabled: true', and 'Root device type: ebs'. A 'Description' section follows, stating 'Microsoft Windows 2022 Datacenter edition. [English]' and 'Microsoft Windows Server 2022 Full Locale English AMI provided by Amazon'. Below the description is a table with columns: Architecture (64-bit (x86)), AMI ID (ami-0f7a0c94dce9ab456), Publish Date (2026-02-12), Username (Administrator), and a 'Verified provider' badge. On the right side of the page is a 'Summary' section with a 'Number of instances' input field set to '1'. It lists the 'Software Image (AMI)' as 'Microsoft Windows Server 2022 ...read more', the 'Virtual server type (instance type)' as 't3.micro', the 'Firewall (security group)' as 'New security group', and the 'Storage (volumes)' as '1 volume(s) - 30 GiB'. At the bottom of the summary are 'Cancel' and 'Launch instance' buttons, along with a 'Preview code' link. The footer of the console shows 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for 2026.

The screenshot shows the 'Instances' page in the AWS Management Console. The top navigation bar is the same as the previous screenshot. The breadcrumb trail is 'EC2 > Instances'. The left sidebar contains a navigation menu with categories: 'EC2' (Dashboard, AWS Global View, Events), 'Instances' (Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager), 'Images' (AMIs, AMI Catalog), and 'Elastic Block Store' (Volumes, Snapshots). The main content area is titled 'Instances' and includes an 'Info' tab. There is a search bar 'Find Instance by attribute or tag (case-sensitive)' and a filter dropdown 'All states'. Below this is a table with columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. The table is currently empty, with a message 'No instances' and 'You do not have any instances in this region'. A 'Launch instances' button is visible. Below the table is a 'Select an instance' section. The footer of the console is the same as the previous screenshot.



Execution Tasks:

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0cb52d1ebb0fad104)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
Manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Manage detailed monitoring
Create Load Balancer
Create AWS budget
Manage CloudWatch alarms

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EC2 > Instances

Instances (1/2) Info
Last updated less than a minute ago
[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D
☒	Project 1	i-0cb52d1ebb0fad104	Running	t3.micro	3/3 checks passed	...	us-east-1c	ec2-54-242-98
☐	Project 2	i-0813ccc1fc44a9dd0	Running	t3.micro	...	...	us-east-1c	ec2-54-226-62

i-0cb52d1ebb0fad104 (Project 1)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0cb52d1ebb0fad104	Public IPv4 address 54.242.98.204 open address	Private IPv4 addresses 172.31.22.200
IPv6 address -	Instance state Running	Public DNS ec2-54-242-98-204.compute-1.amazonaws.com open address
Hostnames	Private IP DNS name (IPv4 only)	

